

Abin Acharya

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Research Interests

Natural Language Processing, Reinforcement Learning, Large Language Models(LLMs), Multi-Agent Systems, Text Summarization, Explainable AI, Adaptive Learning

Education

University of Colorado Colorado Springs (UCCS)

Aug 2025 – Present

Ph.D. in Computer Science

- GPA: 3.92/4.00
- Research Focus: Text Summarization, LLM based Agents, Reinforcement Learning for Reasoning and Optimization
- Graduate Dean's Doctoral Award Recipient
- Relevant Coursework: Artificial Neural Network, Reinforcement Learning, Operating Systems, Database Design, Algorithmic Game Theory and Multi-Agent Systems

Institute of Engineering, Tribhuvan University

Aug 2018 – May 2023

Bachelor in Computer Engineering

- GPA: 3.83/4.00
- Thesis: *Virtual Try-on Using Neural Networks*

Professional Experience

Graduate Teaching Assistant, Department of Computer Science, UCCS

Aug 2025 – Present

- Served as GTA for C#/.NET, Introductory Software Engineering, JAVA and Advanced Software Engineering courses.
- Support course instruction through grading, assignment/project evaluation, and student technical support, providing guidance on programming and software engineering concepts.

Openstack and Proxmox Student Research Assistant UCCS Office of Information Technology

May 2026 – Aug 2026

- Assisted in creating and maintaining clear, comprehensive documentation for Openstack and Proxmox configurations, deployment steps, and testing procedures.
- Organized and formatted technical content for internal reports and public facing documentation websites.
- Supported configuration tasks under supervision, including basic setup and environment replication.

IT Assistant Gurukul College, Chitwan

Sep 2023 – May 2025

- Maintained and managed school IT systems to support smooth daily operations.
- Provided technical support to teachers and staff, resolving software and hardware issues.
- Assisted in implementing and supporting IT-based teaching tools.

Mentoring Experience

Research Mentor, University of Colorado Colorado Springs

Summer 2026

- Mentored undergraduate interns in the NSF-funded Research Experiences for Undergraduates (REU) program on Multi-Agent Systems for Text Summarization
- Provided regular feedback on research progress, presentations, and manuscript preparation.

Publications and Submissions

Conference Papers

- **A. Acharya**, P. Ngo, and J. Kalita, “Varifi: Agentic reinforcement learning for efficient long document summarization via learned chunk selection,” **[Under Review at AAAI-27]**
- P. Ngo, **A. Acharya**, and J. Kalita, “Rafts: Reinforcement-learned agentic feedback for long text summarization,” **Submitted for publication, 2026.**

Journal Papers

- M. A. Mersha, H. Thakur, **A. Acharya**, H. Khatiwada, A. A. Hailu, and J. Kalita, “Multi-agent systems: A survey of explainability, failure analysis, debugging, recovery, and future directions,” **[Under Review at Knowledge Based Systems]**

Talks and Presentations

- “Using Reinforcement Learning for Adaptive Tutoring with LLM-Augmented Decision Making,” presented at **UCCS AI Summit**, Colorado Springs, Colorado, USA, 2026.

Research Projects

- **Adaptive Reinforcement Learning for skill-level Intelligent Tutoring System in Algebra Education**
Developed an RL-based Intelligent Tutoring System using the ASSISTments algebra dataset, modeling student knowledge, boredom, and confidence; benchmarked six tutoring policies, including DQN/DDQN and LLM-based approaches, and developed a hybrid agent achieving $4.6\times$ higher minimum mastery and $13\times$ lower gain variance than DDQN.
- **Explainable AI for malware prediction**
Developed explainable malware prediction models using CNN and TabTransformer on the Microsoft Malware Prediction dataset, applying BOOLXAI to generate human-readable decision rules and analyze feature behavior for more transparent and trustworthy ML.
- **Virtual try on using Neural networks**
Developed a GAN-based Virtual Try-On system using a two-stage GMM-TOM pipeline with MediaPipe, Thin Plate Spline (TPS), U-Net, and VGG-19, achieving 8% NMSE on a 2,000-image test set with realistic garment overlays in 30 seconds per image.
- **Recommendation based discussion forum**
Developed a full-stack Django discussion forum with a content-based recommendation system using cosine similarity to recommend relevant discussions based on user interests, along with personalized search, topic discovery, upvoting, and activity notifications using Python, JavaScript, Bootstrap, and SQLite.

Technical Skills

Programming Languages: Python, C++, C#/.NET, JAVA

Frameworks and Tools: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers.

Data and Visualization: NumPy, Pandas, Matplotlib

Tools and Platforms: Git/GitHub, Docker, Linux/Unix, Google Colab, Jupyter, LaTeX, VS Code, Anaconda

Web Development: Django, PHP, JavaScript, RESTful APIs, SQLite, MySQL

Machine Learning Topics: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI, Context Engineering, Vision Language Models

Professional Service

Session Chair, Research Experiences for Undergraduates (REU) program Summer 2026

- Served as session chair during Mid-term Presentations of REU program.

Awards and Honors

- Graduate Dean's Doctoral Award, University of Colorado Colorado Springs (UCCS)
- Full Scholarship Recipient, UCCS Ph.D. Program
- Full Scholarship Recipient, Institute of Engineering (IOE), Tribhuvan University
- Mahatma Gandhi Scholarship (2016-2018)
- Licensed Professional Engineer, Nepal Engineering Council (NEC)

Certifications

- Communicating Research Findings – CITI Program (May 2026)

Languages

English (Fluent), Nepali (Native)